

COMICCRAFT

DOCUMENT 18: PROPOSED SOLUTION (PROJECT PROPOSAL)

Project Overview

FIELD	DETAILS
Objective	To create an AI-powered web application that enables anyone to generate complete 5-panel comic stories from a simple text prompt, eliminating the need for artistic skills or expensive software
Scope	Full-stack web application integrating Google Gemini models for story generation, Stable Diffusion / FLUX for image generation, and FPDF2 for PDF export. Covers prompt input → AI processing → comic preview → PDF download

Problem Statement

FIELD	DETAILS
Description	Creating comics manually requires artistic talent, expensive tools (Photoshop/Procreate), significant time investment, and knowledge of panel layout and storytelling structure. This barriers prevents many creative individuals from bringing their stories to life
Impact	Solving this will democratize comic creation, enabling writers, students, hobbyists, and content creators to produce professional-looking comics in minutes. Reduces time from hours/days to minutes, eliminates cost barriers, and empowers creative expression

Proposed Solution

FIELD	DETAILS
Approach	Leverage Google Gemini 1.5 Flash for structured outline generation (fast, cost-effective) + Gemini 1.5 Pro for creative narration (high-quality storytelling) + Stable Diffusion / FLUX for visual illustration + FPDF2 for professional PDF output
Key Features	• One-click comic generation from text prompt • 5-panel structured comic with titles, narration, dialogue • AI-generated panel illustrations in multiple art styles • Interactive web preview with panel navigation • Professional multi-page PDF export • Works with or without API keys (mock fallback)

Resource Requirements

RESOURCE TYPE	DESCRIPTION	SPECIFICATION/ALLOCATION
Hardware - Computing	CPU/GPU for local development and inference	NVIDIA GTX 1650 (4GB VRAM) or higher for local image generation
Hardware - Memory	RAM for application and model loading	8 GB minimum, 16 GB recommended
Hardware - Storage	Disk space for code, models, generated files	500 MB for code + 2 GB for cached AI models

Software - Frameworks	Web framework, AI libraries	FastAPI, Uvicorn, Hugging Face Diffusers, FPDF2
Software - Libraries	Additional Python dependencies	Pillow, python-dotenv, google-genai, torch, diffusers
Software - Environment	Development IDE	VS Code, PyCharm, or any Python IDE
Data - Source	User-provided prompts (no training dataset needed)	N/A - zero-shot generation via pre-trained models
Data - Size/Format	User input: text (max 2000 chars)	JSON request body via HTTP POST